

Cybersecurity Concepts Lab for Teachers

Trust your hackles. Pause · Check · Tell.

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 11: Malware Metaphors Without Malware (The Trojan Horse Sort Game)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE	
Big idea	"Malware" is any software made to do harm. It is useful to understand it with three questions: How does it get in? What does it pretend to be? What does it do? The defense for each is different.
Grade bands	K-2 (the trojan horse story and "who gets in the gate"), 3-5 (sort game with simple cards), 6-8 (full sort + three questions), 9-12 (full + blended threats, permissions, supply chain)
CSTA-style concepts	Networks and the Internet (cybersecurity: malware and its effects), Impacts of Computing (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed cards, floor tape for zones, a timer. No devices. No malware, no code, no downloads.
Teachers do it first	Yes. Part A about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Ask the three questions about this fictional card: "A free homework-helper app also quietly records everything typed on the device." How does it get in? What does it pretend to be? What does it do? (No real software is involved in this lesson.)

01 The big idea

The word **malware** (malicious software) covers many kinds of harmful programs. People often say "virus" for all of them, but the kinds act differently, arrive differently and need different defenses. A fun, safe way to learn the differences is with **metaphors and stories**: a wooden horse, a flu, a worm in an apple, a spy, a hostage-taker, an army of puppets. We never need real malware, real code or real files to learn the ideas, and **students never see or handle either**.

The **three-question model** (use this poster all year):

1. **How does it get in?** (Delivery)
2. **What does it pretend to be?** (Disguise)
3. **What does it do once inside?** (Payload)

The defenses mostly stop the first question: don't open the gate to the wrong visitor.

02 What a teacher should know (plain language)

2.1 THE KINDS, IN PLAIN WORDS

KIND	PLAIN DEFINITION	METAPHOR	KEY IDEA
Virus	Attaches itself to a program or file and spreads when that file is run or shared.	A cold that rides on a sick person's handshake	Needs a <i>host</i> and usually a human action to spread.
Worm	Copies itself across networks on its own, without anyone opening anything, often through unpatched holes (Demo 7).	A chain letter that mails itself	Spreads by itself .
Trojan	Pretends to be something useful or fun to get you to run it; it does not spread by itself.	The Trojan Horse	Works through disguise ; the human opens the gate.
Ransomware	Locks your files and demands payment.	A burglar who locks your toy box and wants a fee	See Demo 10.
Spyware (including keyloggers)	Secretly watches and sends information: keystrokes, screens, messages.	A spy in the room	The harm is being watched .
Stalkerware	Spyware installed to secretly monitor a particular person, often by someone they know.	An unwanted tracker in your backpack	Serious personal safety issue; seek help from a trusted adult or a helpline.
Adware / PUP (potentially unwanted program)	Floods you with ads or sneaks in bundled with other downloads.	Door-to-door salespeople who won't leave	Often arrives bundled with "free" apps.
Bot / botnet	Your device is quietly taken over and joins a crowd of devices controlled by someone else.	A puppet army	Your device is used to attack others or send spam.
Rootkit	Hides other bad programs.	A cloak of invisibility	Hard to detect; defenders use special tools.
Cryptominer	Secretly uses your device's power to earn coins for someone else.	A stranger using your electricity	Hot, slow, high battery drain.
Scareware / fake antivirus	Loud pop-up claims you are infected and demands payment or a download.	A fake "firefighter" shouting about a fire	Pressure and fear (Demo 3).

Real incidents blend these. A phishing email (Demo 3) delivers a Trojan, which installs a keylogger and a ransomware tool. Real attackers also steal data and sell access. So the categories are *teaching tools*, not neat boxes. [VERIFY: current industry threat reports, such as Verizon DBIR or ENISA Threat Landscape.]

2.2 WHAT IS NOT MALWARE

- **Phishing** is a trick (Demo 3). It may *deliver* malware, but the message itself is not software.
- **A bug** is a mistake, not an attack (Demo 7).
- **A crash after an update** is usually a bug or compatibility issue.
- **Parental-control or school-management software** installed openly with the knowledge of the device user is monitoring software, not

malware; but secret monitoring of another person is spyware/stalkerware. **Consent and transparency matter.**

- **A real security tool's notification** ("threat blocked") is the defense working.

2.3 HOW MALWARE GETS IN (DELIVERY ROUTES) AND THE MATCHING DEFENSES

ROUTE	EXAMPLE (CONCEPTUAL)	DEFENSE
D1 Disguised download	Free games, "cracked" software, fake "cheats", fake app stores	Install only from official stores and the maker's own site; ask a trusted adult; skip piracy.
D2 Message with a lure	Email attachment, link in a text or chat (Demo 3)	The Six P's; do not open the unexpected; verify.
D3 Shared or found media	An unknown USB stick, a file from a friend's infected computer	Never plug in found devices; scan shared files; keep security tools on.
D4 Network hole	An old unpatched device exposed to the network	Automatic updates (Demo 7), a firewall, replace end-of-life devices.
D5 Over-trusted extension or app permission	A browser extension or app that asks for far more access than needed	Check reviews, publisher and permissions; fewer extensions; remove unused apps.

2.4 GENERAL DEFENSES (DEFENSE IN DEPTH)

- **Updates** on everything (Demo 7).
- **Built-in security tools** (for example, Windows Security or the equivalent on other systems), left on and updated. Antivirus is **a layer**, not a force field: new threats can slip through [VERIFY independent test results].
- **Least privilege**: use a standard (non-administrator) account for daily use, and give apps only the permissions they need.
- **Backups** (Demo 10) so that you can recover.
- **Two locks** (Demo 6) so that stolen passwords are less useful.
- **Careful with money, codes and secrets** (Demo 8).
- **Report and tell**: the earliest report is the best.

2.5 IF A DEVICE MAY BE INFECTED

(Adult-facing, simple version.) Disconnect from the network, stop entering passwords on it, tell the responsible adult (at school: IT), use a different clean device to change important passwords, and follow professional guidance (restore from backup, run trusted scans, reinstall). **Never pay a "tech support" caller or "scareware" pop-up.**

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Everything bad is a virus."	"Virus" is one kind of malware.
"Macs and phones can't get malware."	Less common, but possible, especially via fake apps and lures.
"Only shady sites carry malware."	Compromised or ad-supported legitimate sites can be abused too.
"Antivirus makes me safe."	It helps; judgment and updates matter more.
"I'd notice if I had malware."	Much is silent.

03 Materials (zero to low cost)

- **Card deck** (from the student handout): 20 scenario cards, cut apart. Make 2 to 4 decks for teams.
- **Zone signs** (Virus, Worm, Trojan, Ransomware, Spyware, Adware, Botnet, "Something else", "Not malware") taped to the floor or walls; **route flags** D1 to D5; masking tape and a doorway or tape line for the Gatekeeper's gate; a bell or timer; a cardboard "wooden horse" (optional).
- Student handout. Optional: a "wooden horse" cardboard cutout.

04 The Trojan Horse Sort

4.1 THE SCENARIO CARDS (ALL FICTIONAL; NO CODE, NO REAL FILES)

THE 20 SCENARIO CARDS are printed on the student handout as plain text (no kind or route shown), so you can copy and cut them as they are. The main kind and route for each, and which cards are intentionally ambiguous, are on the teacher answer page. Accept any answer with a reason that uses the three questions; the goal is reasoning.

4.2 GAME 1: THE RELAY SORT (MOVEMENT)

1. Teams line up at one end. Zone signs are at the other end.
2. A runner takes a card, reads it aloud to the team, the team decides in 10 seconds, the runner places the card in a zone, and tags the next runner.
3. After all 20 cards, the teacher reveals the answers. Teams score 2 points per correct zone, 1 point if their reason is correct for an ambiguous card.
4. Debrief: which were hardest? Why?

4.3 GAME 2: THE GATEKEEPER (DEFENSE DECISIONS)

A student is the **Gatekeeper** at the gate. Visitors (students with disguise cards) approach: "Free game!", "Update now!", "Important invoice!", "Unknown USB!", "Your friend's file!", "Official app store app!" The Gatekeeper uses a **Check Card** (Source? Needed? Expected? Permissions?) to decide: *Open the gate / Hold and ask an adult / Close the gate*. The class votes. After each decision, the teacher reveals what was inside. Students see that **the same disguise can hide different things** and that the best defense is a routine.

4.4 GAME 3: THE WOODEN HORSE STORY (K-5 VERSION)

Read aloud: "Long ago a city was surrounded. The enemy left a huge wooden horse outside the gate as a 'gift'. The city opened the gate. Inside the horse hid soldiers. In computers, a Trojan is a program that looks like a gift but hides something else. Who opens the gate? You do! So you are the gate keeper." Act it out with a "horse" made of chairs and a "gift" label. Then: "What would a smart gatekeeper ask?" (Who gave it? Was I expecting it? Can a grown-up check?) Link to Pause · Check · Tell.

05 Part A: Teachers do it first (about 20 minutes)

1. **Quick sort (8 min).** In groups, sort the 20 cards into zones without the answers. Notice your disagreements.
2. **Reveal and reason (5 min).** Compare with the answer page. Which of your reasons used the three questions?
3. **Gatekeeper (5 min).** Play one round with colleagues as visitors. What feels natural (opening) and what needs a checklist?
4. **Reflect (2 min).** Which card felt closest to your own habits?

06 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-3	Hook with the Wooden Horse story and the three-question poster.
3-10	Relay Sort with 10 cards (cards 1, 2, 3, 4, 6, 7, 11, 12, 15, 20).
10-13	Reveal; discuss the "gate": how did each get in?
13-15	Exit slip: "Before I open the gate, I ask ____."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and vocabulary (delivery, disguise, payload).
5-20	Relay Sort with all 20 cards, two rounds (by kind, then by route).
20-32	Gatekeeper game with Check Cards, three rounds, student Gatekeepers.
32-40	Defense in depth: students build a "castle" poster with layers (updates, two locks, backups, built-in security, ask first).
40-45	Exit ticket and handout.

07 Teacher script

Hook (2 minutes). "A long time ago, a city got a present: a giant wooden horse. They brought it inside. Surprise! It was full of soldiers. In computers a 'Trojan' is a program that looks like a gift but hides something else. There's a whole zoo of harmful programs, but I want you to ask three questions about any of them: How does it get in? What does it pretend to be? What does it do? When you know those answers, you know where to put your guard."

During the sort. "Card 3: it spreads without anyone clicking. That's the big difference from a Trojan, where a human opens the gate. Card 11: Is a fake bank message software? No. It's a trick. But it can carry a link to something that is."

Gatekeeper. "Our gatekeeper doesn't need to know every type of monster. The routine works: Where did it come from? Did I expect it? Does it ask for too much? If I'm unsure, I hold it at the gate and ask an adult."

Close. "Real defenders keep several layers: updates, two locks, backups, built-in security tools, and a careful gatekeeper. If you ever open something and get a funny feeling, pause and tell a trusted adult. It's never your fault, and the sooner we know, the better."

08 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The wooden horse and the gate. A gift that isn't what it looks like. Practice "Ask first": "Can you check this with me?" No names of malware types.
3-5	The zoo of sneaky guests. Simple sort of 8 cards into Trojan, ad-annoy, spy, not-a-problem. Use the Gatekeeper game.
6-8	Three questions and the full sort. Compare virus, worm, Trojan. Discuss consent and monitoring versus spyware.
9-12	Kill chain thinking and defense in depth. Delivery, execution, persistence, command-and-control and impact as a concept (no technical detail); blended threats; supply-chain risk; permission models; incident response (Demo 13).

09 Discussion questions

1. Which defense stops the *most* kinds? (Updates, careful gatekeeping, backups.)
2. Why does a worm need no human help, but a Trojan does?
3. Why can the same free download be harmless on one site and harmful on another?
4. (9-12) How does the three-question model map onto a real incident story you've read?

10 Level-up challenge

- **Build your own deck.** Groups write 5 new fictional cards (no code, no brand names) with a three-question answer; another group sorts them.
- **Permission audit.** With a trusted adult, on their own device, students check one app's permissions against its purpose and report the *number*, not the details.
- **Case study (9-12).** Read a public advisory about a ransomware or malware incident (for example from CISA) and map it to the three questions [VERIFY a suitable public case].

11 Safety and ethics notes (defense only)

- **No real malware, ever.** Do not bring in, download, display, or discuss how to create malware. Do not show screenshots of live malicious pages. Cards are fictional descriptions.
- **No code or "how it works under the hood" detail beyond the plain definitions here.** If a student asks "how do I write one?" say: "We're not going to build one here. If you love this, there are legal ways to learn: ethical hacking clubs, capture-the-flag competitions with permission, and computer science classes. Let's talk about that in Demo 14."
- **No testing on any real device** or network, and no plugging unknown media into school devices.
- **Stalkerware is a safety issue.** If a student suspects a device is being secretly monitored by someone they know, do not investigate in class; follow the school's reporting process and involve the counselor (see the Index box). Do not advise the student to confront the person or to remove or change anything on the device, because that can alert the person and raise the risk; the counselor and specialist helplines can plan a safe response.
- **Be careful with fear.** Keep it factual and calm.

12 Assessment idea

- **Exit ticket.** "Name one kind of malware, how it gets in, and one defense."
Rubric: 3 = all three accurate; 2 = two; 1 = "viruses are bad".
- **Gatekeeper rubric.** Observe students in Game 2. Do they ask about source, expectation and permissions? Do they know whom to ask?

13 Sources and [VERIFY] list

- NIST glossary and guidance on malware (NIST SP 800-83 Rev. 1 "Guide to Malware Incident Prevention and Handling for Desktops and Laptops", July 2013) [VERIFIED 2026-10-05: csrc.nist.gov].
- CISA "Protecting Against Malware" tips [not checked].
- Coalition Against Stalkerware resources [VERIFY].
- Verizon DBIR / ENISA Threat Landscape for current delivery routes [VERIFY].
- CSTA 3A-NI-05 on how sensitive data can be affected by malware [VERIFIED 2026-10-05: CSTA K-12 CS Standards (2017, rev.) text read; mapping approximate].
- Independent antivirus test labs for limits of antivirus products (AV-TEST, AV-Comparatives) [VERIFY].

Student Handout: The Trojan Horse Sort

Name: _____

Three questions for any sneaky program

1. How does it **get in**?
2. What does it **pretend** to be?
3. What does it **do**?

Match the kind to the clue

WRITE THE LETTER OF the kind. Kinds: A Worm, B Spyware, C Trojan, D Ransomware.

1. Looks like a gift but hides something else ____
2. Copies itself across a network with no clicking ____
3. Quietly watches what you type ____
4. Locks your files and demands money ____

The Gate

Five ways in: **D1** disguised download, **D2** message with a lure, **D3** shared or found media, **D4** network hole (no updates), **D5** over-trusting extension or app.

Which way in was each? Write D1 to D5.

- A free "cheat tool": ____
- A message with an "invoice" attachment: ____
- A USB stick found on the ground: ____
- An old unupdated computer on a network: ____

Gatekeeper checklist

Before I open the gate:

- ☐ **Source:** who gave it to me?
- ☐ **Expected:** was I waiting for it?
- ☐ **Permissions:** does it ask for too much?
- ☐ **Ask:** have I asked a trusted adult?

Not malware!

EACH OF THESE IS NOT malware. Write why for each: **a fake bank message**
_____ / **a game crash after an update** _____ / **a built-in "threat**
blocked" message _____

Remember

- Don't plug in found USB sticks. Install from official places. Ask first.
- If you opened something and your hackles go up, **Pause. Tell.** It's **not your fault.**

Scenario Cards (all made up; cut apart)

1. A free homework-helper app also quietly records everything typed on the device.
2. A program hides inside a shared document and copies itself into other documents whenever someone opens it.
3. A program on one computer finds other unpatched computers on the school network and copies itself with no one clicking anything.
4. A screen says your photos are locked and you must pay or lose them.
5. Hundreds of ordinary computers secretly take orders from one remote controller.
6. A cool-looking "game cheat" tool does something completely different when you run it.
7. After installing a free app, pop-up ads appear everywhere, even on the home screen.
8. A loud pop-up says "You have 47 viruses! Pay now!"
9. Your laptop is hot and slow because it is secretly doing work for someone else's benefit.
10. A tool that hides other harmful programs from the list of running programs.
11. A message pretends to be your bank and asks for your password.
12. A game crashes after an update.
13. A parent installs a monitoring app on a child's tablet, and the child knows and has agreed to the rules.
14. An app secretly installed on someone's phone tracks their location and messages without their knowledge.
15. An email attachment called "invoice" installs a program when you open it.
16. A USB stick found in a parking lot is plugged into a computer "just to see".
17. A helpful browser extension is sold to a new owner and starts injecting ads and tracking.
18. A text says "Your parcel is waiting!" and the link installs an app that can read your messages.

19. A program copies itself to every USB drive plugged into the computer.
20. A message from the device's built-in security tool says a threat was blocked.

End of student handout.

SECTION

3

3

TEACHER COPY ONLY: Answers for Demo 11 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Handout match: 1 C (Trojan), 2 A (Worm), 3 B (Spyware), 4 D (Ransomware).

Gate: free cheat tool D1; invoice attachment D2; found USB D3; old unupdated computer D4. **Not malware:** the fake bank message is phishing, a trick, not software; the crash is a bug; the built-in message is the defense working.

Scenario cards: kind and route

#	MAIN KIND	ROUTE
1	Spyware (keylogger)	D1
2	Virus	D3
3	Worm	D4
4	Ransomware	D2 (or others)
5	Botnet	D4 (bots are recruited through many routes)
6	Trojan	D1
7	Adware	D1
8	Scareware	D1 (a lure on a web page that pushes a "fix" download)
9	Cryptominer	D1 or D5
10	Rootkit	D4 or D1
11	Not malware (phishing)	D2

3

#	MAIN KIND	ROUTE
12	Not malware (a bug)	none
13	Not malware (monitoring with knowledge and consent); discuss transparency	none
14	Spyware (stalkerware), installed by someone with access to the phone	D1 (see the stalkerware safety note)
15	Trojan	D2
16	Delivery risk (could carry anything)	D3
17	Adware or spyware	D5
18	Trojan or spyware	D2
19	Worm or virus (spreads via media)	D3
20	Not malware (a defense working)	none

Intentionally ambiguous (accept any answer with a reason using the three questions): cards 8, 10, 13, 16, 17, 18 and 19. Score: 2 points per correct zone, 1 point if the reason is right for an ambiguous card.